

Infantile-Onset Multisystem Neurologic, Endocrine, and Pancreatic Disease (IMNEPD1) — Comprehensive Research Report

Disease: Infantile-Onset Multisystem Neurologic, Endocrine, and Pancreatic Disease, type 1 (IMNEPD1) **MONDO ID:** MONDO:0024189 | **OMIM:** #616263 | **ORPHA:** 456312 | **Gene:** *PTRH2* (HGNC:24265) **Category:** Mendelian (autosomal recessive)

Summary

Infantile-onset multisystem neurologic, endocrine, and pancreatic disease type 1 (IMNEPD1) is an ultra-rare autosomal-recessive Mendelian disorder caused by biallelic loss- or reduction-of-function variants in *PTRH2* (peptidyl-tRNA hydrolase 2, also called BIT1; chromosome 17q23.1, UniProt Q9Y3E5). First delineated by Hu and colleagues in 2014 in two siblings of consanguineous Turkish parents, the disorder has since been reported in roughly 19–32 patients worldwide, almost exclusively from consanguineous Middle Eastern and North African families. The recurrent missense allele **p.Gln85Pro (Q85P; c.254A>C)** accounts for approximately half of all reported cases and behaves as a founder/hotspot allele across Tunisian, Saudi, and other Arab communities.

Clinically, IMNEPD1 is a **neuro-predominant multisystem atrophy** presenting in infancy with global developmental delay/intellectual disability, ataxia with progressive cerebellar atrophy/hypoplasia, demyelinating sensorimotor peripheral neuropathy, distal muscle weakness, and sensorineural hearing loss, together with a variable endocrine-pancreatic component: exocrine pancreatic insufficiency (~33%), diabetes mellitus (~40%), hypothyroidism (~22%), and liver abnormality (~17%). Additional features include postnatal microcephaly, craniofacial dysmorphism, and hand deformity.

Mechanistically, PTRH2/BIT1 is a bifunctional mitochondrial protein. It performs canonical peptidyl-tRNA hydrolase activity in mitochondrial translation quality control, and it "moonlights" as an adhesion-regulated survival switch: when cells are attached to extracellular matrix, PTRH2 supports FAK→PI3K/AKT→NF-κB→Bcl-2 survival signaling and mTOR activity while restraining ERK; upon matrix detachment it is released to the cytosol to trigger caspase-independent anoikis. Newer work adds a third arm — regulation of mitochondrial dynamics through mitofusins MFN1/2 and FKBP8, and control of Complex I subunit mt-ND5 via the deubiquitinase TRABID. Biallelic PTRH2 loss converges on impaired survival of post-mitotic cells (Purkinje neurons, peripheral nerve, myocytes, pancreatic acini, hepatocytes), producing progressive cellular atrophy. Diagnosis is molecular (WES/WGS); there is no disease-modifying therapy, and management is entirely supportive and multidisciplinary. Constitutive and Purkinje-cell-conditional *Ptrh2*-knockout mice faithfully recapitulate the human phenotype.

Key Findings

F001 — IMNEPD1 is caused by biallelic *PTRH2* mutations (autosomal recessive)

IMNEPD1 (MONDO:0024189; OMIM #616263) is caused by homozygous/biallelic mutations in *PTRH2* (peptidyl-tRNA hydrolase 2; HGNC:24265; OMIM 608625; *chromosome 17q23.1*). The disorder was originally described by Hu et al. 2014 (Ann Clin Transl Neurol; [PMID: 25574476](#)) in two siblings of consanguineous Turkish parents carrying a homozygous frameshift variant *c.269_270delCT (p.Ala90fs)**, identified by homozygosity mapping plus whole-exome sequencing and segregating with disease. Inheritance is autosomal recessive with variable expressivity, and nearly all reported families are consanguineous. This establishes IMNEPD1 as a monogenic Mendelian disease with a single causal gene.

F002 — Core phenotype and frequencies across a ~32-case cohort

Updated cohort meta-analyses (Sharkia et al., *Genes* 2023, PMC10217894; 2024 update PMC11675358) across roughly 32 reported patients yield the following phenotype frequencies:

Phenotype	Frequency	Suggested HPO term
Motor delay	~94%	HP:0001270 (Motor delay)
Peripheral neuropathy	~89%	HP:0009830 (Peripheral neuropathy)
Intellectual disability	87.5%	HP:0001249 (Intellectual disability)
Distal muscle weakness	85.2%	HP:0002460 (Distal muscle weakness)
Sensorineural hearing impairment	~78%	HP:0000407 (Sensorineural hearing impairment)
Ataxia	~77%	HP:0001251 (Ataxia)
Hand deformity	~58%	HP:0001155 (Abnormality of the hand)
Cerebellar atrophy/hypoplasia	~56.5%	HP:0001272 / HP:0001321
Craniofacial dysmorphism	~53%	HP:0001999 (Abnormal facial shape)
Diabetes mellitus	~40%	HP:0000819 (Diabetes mellitus)
Exocrine pancreatic abnormality	~33%	HP:0001738 (Exocrine pancreatic insufficiency)
Hypothyroidism	~22%	HP:0000821 (Hypothyroidism)
Liver abnormality	~17%	HP:0001392 (Abnormality of the liver)

The OMIM-emphasized **core tetrad** is global developmental delay/intellectual disability with speech delay, ataxia, sensorineural hearing loss, and exocrine pancreatic insufficiency. Additional recurrent features include postnatal microcephaly, peripheral demyelinating sensorimotor neuropathy, cerebellar atrophy, and dysmorphism (midface hypoplasia, thin upper lip, exotropia, ptosis). All phenotypes have onset in infancy/early childhood and are progressive, with severe impact on daily functioning, mobility, communication, and nutrition.

F003 — PTRH2/BIT1 is an adhesion-regulated survival vs. anoikis switch; mTOR loss drives Purkinje-cell atrophy

PTRH2/BIT1 is a bifunctional mitochondrial protein with (1) canonical **peptidyl-tRNA hydrolase activity** (it releases the peptidyl moiety from tRNA, preventing toxic accumulation of peptidyl-tRNA during mitochondrial translation) and (2) hydrolase-independent **"moonlighting" signaling**. In ECM-attached cells, PTRH2 complexes with focal adhesion kinase (FAK) at the membrane, driving PI3K/AKT/NF- κ B signaling and Bcl-2 transcription to

promote cell survival (Griffiths 2011). On matrix detachment, mitochondrial BIT1 is released into the cytoplasm, binds the Groucho/TLE corepressor AES, and induces caspase-independent apoptosis (anoikis; Jan et al. 2004, PMID: 15006356). PTRH2 negatively regulates ERK — Bit1-null MEFs are anoikis-resistant with increased phospho-ERK. In IMNEPD patient cells and *Ptrh2* mutant mouse brain there is decreased FAK/mTOR activation and increased phospho-ERK. The Purkinje-cell-specific knockout (*Ptrh2*^{ΔPC}) shows reduced ribosomal protein S6 phosphorylation (an mTOR pathway readout), stunted dendrites, and Purkinje-cell atrophy (*Cerebellum* 2022, doi:10.1007/s12311-022-01488-z). In muscle, PTRH2 complexes with α7β1 integrin; its loss decreases α7 integrin and integrin signaling, phenocopying α7-integrin-null myopathy.

F004 — Genotype–phenotype correlation and mouse-model recapitulation

The recurrent variant **Q85P (c.254A>C, p.Gln85Pro)** accounts for ~50% of cases and is shared across Tunisian, Saudi, and other Arab consanguineous communities. It is a destabilizing missense change: patient fibroblasts show normal *PTRH2* mRNA but strongly reduced protein (Picker-Minh et al. 2016, PMID: 27129381). A broad correlation emerges: **missense variants (Q85P) tend to produce milder "core" phenotypes, whereas nonsense/truncating variants** (e.g., p.Trp108*, p.Ala90fs) associate with a fuller, more severe multisystem phenotype. Even the identical Q85P allele yields variable expressivity — the Sharkia 2017 family showed normal intelligence, milder microcephaly, delayed puberty, myopia, and pain insensitivity.

Ptrh2-null mice recapitulate the human disease: they are born at normal size but show postnatal failure to thrive, ataxia, and muscle weakness, with death by postnatal day 8–14. They develop cerebellar atrophy and an underdeveloped exocrine pancreas (reduced fecal elastase, smaller acini) with preserved islets of Langerhans, plus smaller neurons, myocytes, acinar cells, and hepatocytes. At the signaling level they show reduced phospho-FAK and Bcl-2 and increased phospho-ERK — mirroring the human molecular pathology.

F005 — Variant catalog, allele frequency, and identifiers

Reported IMNEPD1 *PTRH2* variants (transcript NM_016077.5), all germline and biallelic (homozygous in consanguineous families or compound heterozygous):

Variant (cDNA)	Protein	Type	Note / reference
c.254A>C	p.Gln85Pro (Q85P)	Missense	Recurrent hotspot ~50% of cases; Alazami 2015 PMID: 25558065 ; once heterozygous in gnomAD
c.68T>C	p.Val23Ala (V23A)	Missense	First Iranian case
c.254A>G	p.Gln85Arg (Q85R)	Missense	—
c.280T>A	p.Tyr94Asn (Y94N)	Missense	—
c.269_270delCT	p.Ala90fs	Frameshift	Original family; Hu 2014 PMID: 25574476
c.324G>A	p.Trp108*	Nonsense	Le 2019 PMID: 31057140
—	p.Glu110*	Nonsense	—
c.127dupA	p.Ser43Lysfs*11	Frameshift	Parida 2021

All variants are ultra-rare and absent/near-absent from gnomAD, 1000 Genomes, and ExAC, and are classified pathogenic/likely pathogenic per ACMG/AMP criteria. Functional consequence is loss/reduction of function (missense variants destabilize the protein; truncating variants cause loss of function).

Identifiers: MONDO:0024189; OMIM #616263; ORPHA:456312; ICD-10 Q87.8; ICD-11 LD90.Y; UMLS C5779989. **Gene:** *PTRH2*, HGNC:24265, OMIM *608625, 17q23.1, UniProt Q9Y3E5, NCBI Gene 51651.

F006 — PTRH2 regulates mitochondrial fusion; truncation mutants drive fragmentation, rescued by amino-acid deprivation

Na et al., *Mol Med* 2026 ([PMID: 41807994](#)) showed that *PTRH2* interacts with mitofusins *MFN1/2*, interfering with *MFN* dimerization to suppress mitochondrial fusion. The *IMNEPD* truncation mutants *A90fs and W108 show enhanced binding to *MFN1/2*, driving excessive fission/mitochondrial fragmentation and perinuclear aggregation via recruitment of FKBP8, together with impaired mitophagy (mito-Keima assay), reduced ATP and membrane potential, and increased ROS. Critically, the pathological *MFN*–*PTRH2*–FKBP8 interaction is

alleviated by **amino-acid deprivation (EBSS + QLR)**, which promotes fusion and reduces mitochondrial aggregation — a candidate therapeutic rescue condition. Complementary work (*PNAS Nexus* 2025) shows that mitochondrial PTRH2 controls the deubiquitinase **TRABID** to regulate the stability of **mt-ND5** (a Complex I subunit) and hence oxidative metabolism.

F007 — Diagnosis, supportive management, and prognosis

Diagnosis is molecular: there is no specific biochemical marker, so diagnosis relies on WES/WGS (or targeted *PTRH2* sequencing / neuropathy-ataxia gene panels) confirming biallelic *PTRH2* variants. Supportive workup includes brain MRI (progressive cerebellar atrophy/hypoplasia), nerve conduction studies/EMG (sensorimotor, often demyelinating neuropathy), audiometry/BAER (sensorineural hearing loss), fecal pancreatic elastase-1 (reduced → exocrine pancreatic insufficiency), fat-soluble vitamin levels (A/D/E/K deficiency from EPI), thyroid function (hypothyroidism), fasting glucose/HbA1c (diabetes in ~40%), liver enzymes/ultrasound (steatosis/fibrosis), and EEG (some patients have epilepsy).

Management is entirely supportive and multidisciplinary — no disease-modifying or curative therapy exists:

Manifestation	Intervention	Suggested NCIT concept
Exocrine pancreatic insufficiency	Pancreatic enzyme replacement therapy (PERT) + fat-soluble vitamin supplementation	NCIT:C158477 (Pancreatic Enzyme Replacement)
Hypothyroidism	Levothyroxine	NCIT:C29099 (Levothyroxine)
Diabetes mellitus	Insulin	NCIT:C2271 (Insulin)
Sensorineural hearing loss	Hearing aids / cochlear implantation	NCIT:C99277 (Cochlear Implant)
Neuropathy/ataxia/weakness	Physiotherapy, occupational & speech therapy, orthoses	NCIT:C15352 (Physical Therapy)
Seizures	Anticonvulsants	NCIT:C264 (Anticonvulsant Agent)
Failure to thrive	Nutritional support	NCIT:C15417 (Nutritional Support)

Prognosis is chronic, progressive, and lifelong. Severity is variable and partly genotype-dependent (truncating variants more severe). Human survival ranges from severe infantile forms to milder cases reaching adulthood (one patient's gait instability was first noted at ~50 years). No formal survival statistics exist owing to the disease's rarity.

F008 — Comparative biology, orthologs, and digenic PTRH2+KIF1A syndrome

PTRH2/BIT1 is a 179-amino-acid (~27 kDa) protein with an N-terminal mitochondrial localization sequence and a C-terminal UPF0099 (PTH2) domain, conserved from bacteria to human. Orthologs include mouse *Pthr2* (MGI:2444848; the primary disease model), rat *Pthr2* (RGD:1602115), zebrafish *ptrh2* (ZFIN ZDB-GENE-050522-163; predicted PTH activity and anoikis regulation), and the yeast PTH2 ortholog (retains peptidyl-tRNA hydrolase activity but is non-essential; **Rosas-Sandoval et al. 2002**, *PNAS* PMID: 12475932). Bacterial/archaeal PTH enzymes are essential for translational quality control, whereas the eukaryotic PTH2 hydrolase role is non-essential. No naturally occurring *PTRH2* disease has been reported in companion animals or wildlife (OMIA: none), and the disorder is not zoonotic or infectious. A **digenic** finding (**Rea et al. 2021**, PMID: 33717719) describes a syndrome arising from synergistic *PTRH2* + *KIF1A* variants (hereditary axonopathy, outer hair cell dysfunction, intellectual disability, pancreatic lipomatosis, diabetes, cerebellar atrophy, vertebral artery hypoplasia), proposing an umbrella term "neuro-pancreatic syndromes (NPS)" that encompasses IMNEPD.

F009 — Consolidated pathogenesis model

Integrating human case reports/series (n≈19–32), patient fibroblasts, CRISPR cell lines, and both constitutive and Purkinje-conditional *Pthr2* knockout mice, the causal model is: **biallelic PTRH2 loss/destabilizing variant → reduced functional PTRH2/BIT1 → three convergent arms** — (A) impaired mitochondrial peptidyl-tRNA hydrolysis plus PTRH2–TRABID–mt-ND5 Complex I destabilization (OXPHOS/ATP deficit, ROS); (B) loss of adhesion-dependent FAK→PI3K/AKT→NF-κB→Bcl-2 survival signaling with mTOR downregulation (reduced S6) and ERK de-repression; and (C) MFN1/2–FKBP8-driven mitochondrial fragmentation/aggregation with mitophagy failure — **→ reduced growth/survival of post-mitotic cells → progressive atrophy** of Purkinje neurons, peripheral nerves, myocytes, pancreatic acini, and hepatocytes → the full multisystem phenotype.

Section-by-Section Detail

1. Disease Information

A concise overview: IMNEPD1 is an autosomal-recessive multisystem disorder of infancy characterized by a neurologic core (developmental delay/intellectual disability, ataxia, cerebellar atrophy, demyelinating peripheral neuropathy, sensorineural hearing loss) with variable endocrine and pancreatic involvement. **Identifiers:** MONDO:0024189, OMIM #616263, ORPHA:456312, ICD-10 Q87.8, ICD-11 LD90.Y, UMLS C5779989. **Synonyms:** IMNEPD; IMNEPD1; PTRH2-related disorder; PTRH2 deficiency; the disorder falls within the proposed "neuro-pancreatic syndromes (NPS)." Information is derived from **aggregated disease-level resources** (OMIM, Orphanet) built on individual case reports and small consanguineous family series — not from large EHR datasets.

2. Etiology

The **primary cause is genetic**: biallelic (homozygous or compound-heterozygous) loss/reduction-of-function variants in *PTRH2*. There are no established environmental, infectious, or toxic causes. The dominant **genetic risk factor is consanguinity** (autozygosity for the recurrent Q85P founder allele or family-private truncating variants); virtually all families are consanguineous, and geographic clustering is in the Middle East/North Africa. No protective alleles or gene–environment interactions are described. The digenic *PTRH2*+*KIF1A* observation (Rea 2021) suggests a second gene can modify or expand the phenotype in rare instances.

3. Phenotypes

See the frequency table in F002. Phenotype **types** span clinical signs (ataxia, distal weakness, dysmorphism), laboratory abnormalities (reduced fecal elastase, abnormal thyroid function, hyperglycemia, abnormal liver enzymes), and behavioral/developmental changes (intellectual disability, speech delay). **Onset** is neonatal/infantile for most neurologic features; endocrine features (diabetes, hypothyroidism) may emerge later in childhood/adolescence. **Severity** is variable and partly genotype-dependent; **progression** is generally progressive. **Quality-of-life impact** is high — combined motor, cognitive, sensory (hearing), and nutritional impairment.

4. Genetic / Molecular Information

See F005. **Causal gene:** *PTRH2* (HGNC:24265; OMIM 608625). *Variant classes:* missense (destabilizing), nonsense, and frameshift, all germline. *Allele frequency:* ultra-rare/absent in population databases. *Functional consequence:* loss/reduction of function. *Modifier genes:* KIF1A* implicated in a digenic case. No specific epigenetic mechanism or chromosomal abnormality is associated.

5. Environmental Information

Not applicable — IMNEPD1 is a monogenic Mendelian disorder with no established environmental, lifestyle, or infectious contributors. Consanguinity is a demographic/genetic risk determinant rather than an environmental exposure.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram in the Mechanistic Model section below.

7. Anatomical Structures Affected

Primary organs/systems: cerebellum (UBERON:0002037) — especially Purkinje cells; peripheral nervous system (UBERON:0000010); exocrine pancreas (UBERON:0000017); cochlea/inner ear (UBERON:0001844); skeletal muscle (UBERON:0001134). **Secondary:** liver (UBERON:0002107), thyroid gland (UBERON:0002046), endocrine pancreas (variable). **Cell types (CL):** Purkinje cell (CL:0000121), neuron (CL:0000540), Schwann cell (CL:0002573), pancreatic acinar cell (CL:0002064), skeletal muscle fiber (CL:0008002), hepatocyte (CL:0000182), cochlear hair cell (CL:0000855). **Subcellular (GO CC):** mitochondrion (GO:0005739), mitochondrial inner membrane (GO:0005743), focal adhesion (GO:0005925). **Lateralization:** bilateral/symmetric.

8. Temporal Development

Onset: congenital/infantile (postnatal). **Pattern:** chronic, progressive. **Course:** neurologic decline with cerebellar atrophy; endocrine/pancreatic features may appear or worsen over time. **Duration:** lifelong. No remission. The infant/early-childhood window is the critical period for supportive intervention (hearing, nutrition, developmental therapies).

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** high/complete for biallelic pathogenic genotypes, with **variable expressivity** (even within Q85P homozygotes). **Prevalence:** <1/1,000,000 (Orphanet); ~19–32 patients reported. **Founder effect:** Q85P across Middle Eastern/North African consanguineous populations. **Consanguinity:** central. No genetic anticipation or documented germline mosaicism. **Sex ratio:** approximately equal (autosomal). **Carrier frequency:** not established; alleles near-absent in gnomAD.

10. Diagnostics

See F007. Diagnosis is molecular (WES/WGS or targeted *PTRH2*/neuropathy-ataxia panel). Supportive tests: brain MRI, NCS/EMG, audiometry/BAER, fecal elastase-1, fat-soluble vitamins, thyroid function, glucose/HbA1c, liver enzymes/ultrasound, EEG. **Differential diagnosis:** other autosomal-recessive cerebellar ataxias/PCH, CMT/hereditary neuropathies, syndromic sensorineural hearing loss, and Shwachman-Diamond and other exocrine-pancreatic-insufficiency syndromes; molecular testing is discriminating. **Screening:** cascade/carrier testing in affected consanguineous families; prenatal/preimplantation testing where the familial variant is known.

11. Outcome / Prognosis

Chronic, progressive, lifelong; severity partly genotype-dependent (truncating > missense). Morbidity from combined motor, cognitive, sensory, and nutritional deficits is high. No formal survival statistics; outcomes range from severe infantile forms to milder adult survivors. Prognostic factors: variant type (truncating vs missense) and extent of endocrine/pancreatic involvement.

12. Treatment

Supportive/multidisciplinary only (see F007 table). No pharmacotherapy targets the primary defect; no gene, cell, or RNA therapy exists. The amino-acid-deprivation/mitochondrial-fusion rescue (Na 2026) is a preclinical cell-model lead, not a clinical therapy.

13. Prevention

No primary prevention beyond **genetic counseling** for consanguineous families, **carrier/cascade testing**, and **prenatal/preimplantation genetic diagnosis** where the familial variant is known. Secondary/tertiary prevention centers on early detection and management of complications (hearing, nutrition/EPI, endocrine surveillance).

14. Other Species / Natural Disease

See F008. Orthologs in mouse, rat, zebrafish, and yeast; deep conservation of the PTH2 domain. No naturally occurring animal disease (OMIA: none); not zoonotic. NCBI Taxon: *Homo sapiens* (9606); mouse (10090).

15. Model Organisms

The **mouse** is the primary and faithful model: constitutive *Pthr2*-knockout (postnatal failure to thrive, ataxia, muscle weakness, cerebellar atrophy, exocrine pancreatic hypoplasia, early death P8–14) and **Purkinje-cell-conditional knockout** (reduced phospho-S6/mTOR readout, dendritic stunting, Purkinje-cell atrophy). Resources: MGI:2444848. **Limitations:** mice die too early to model late endocrine features (diabetes, hypothyroidism) and relatively spare islets, diverging from the ~40% human diabetes frequency. Zebrafish and yeast provide comparative/orthology systems.

Mechanistic Model / Interpretation

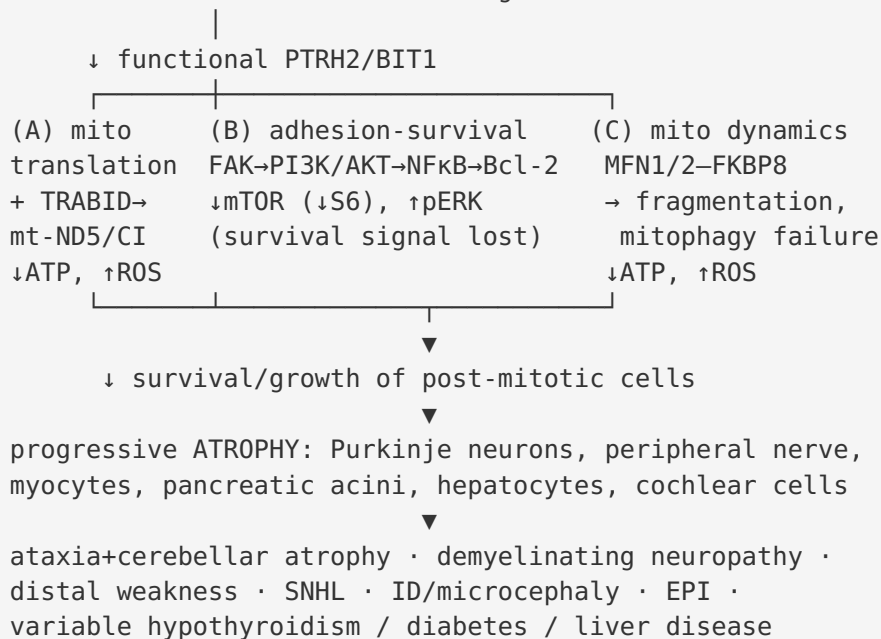
Ordered causal chain (initiating lesion → clinical manifestation)

1. A biallelic loss-of-function or destabilizing missense variant in *PTRH2* (e.g., Q85P protein destabilization; A90fs/W108 *truncation*) *leads to** reduced functional PTRH2/BIT1 protein in mitochondria.
2. Reduced PTRH2 **results in** impaired mitochondrial peptidyl-tRNA hydrolysis and, via loss of PTRH2–TRABID control of mt-ND5, destabilized Complex I → OXPHOS/ATP deficit and increased ROS (demonstrated in cell models; *PNAS Nexus* 2025).
3. In parallel, reduced PTRH2 **leads to** loss of adhesion-dependent FAK→PI3K/AKT→NF-κB→Bcl-2 survival signaling, with downstream mTOR downregulation (reduced phospho-

S6) and de-repression of ERK (demonstrated in patient cells and mouse brain/Purkinje cells).

4. In a third branch, truncation mutants show enhanced MFN1/2 binding and FKBP8 recruitment, which **results in** excessive mitochondrial fission/fragmentation, perinuclear aggregation, and mitophagy failure (demonstrated for A90fs/W108; Mol Med* 2026).
5. These converging deficits **result in** reduced survival and growth of post-mitotic, high-energy-demand cells.
6. Cell-type-specific atrophy **leads to** the clinical phenotype: Purkinje/cerebellum → ataxia + cerebellar atrophy; peripheral nerve → demyelinating sensorimotor neuropathy + distal weakness; cochlear cells → sensorineural hearing loss; CNS neurons → intellectual disability + microcephaly; pancreatic acini → exocrine pancreatic insufficiency (islets relatively spared); hepatocytes/endocrine → variable liver disease, hypothyroidism, diabetes.

PTRH2 biallelic LoF / destabilizing variant



Upstream vs downstream: the mutation and PTRH2 depletion are most upstream; the three mitochondrial/signaling arms are proximal molecular consequences; cellular atrophy is the intermediate cellular phenotype; organ dysfunction is downstream.

Ontology suggestions. GO biological processes: peptidyl-tRNA hydrolase activity (GO:0004045), mitochondrial translation (GO:0032543), regulation of mitochondrial fission (GO:0090140), mitochondrial fusion (GO:0008053), anoikis (GO:0043276), apoptotic process (GO:0006915), mitophagy (GO:0000423). GO cellular components: mitochondrion (GO:0005739), mitochondrial

inner membrane (GO:0005743), focal adhesion (GO:0005925). CL cell types: Purkinje cell (CL:0000121), neuron (CL:0000540), Schwann cell (CL:0002573), pancreatic acinar cell (CL:0002064), skeletal muscle fiber (CL:0008002), hepatocyte (CL:0000182), cochlear hair cell (CL:0000855). UBERON: cerebellum (UBERON:0002037), peripheral nervous system (UBERON:0000010), exocrine pancreas (UBERON:0000017), cochlea (UBERON:0001844), liver (UBERON:0002107), thyroid gland (UBERON:0002046). CHEBI: ATP (CHEBI:15422), reactive oxygen species (CHEBI:26523).

Evidence Base

PMID	First author / year	Contribution	Evidence type
25574476	Hu 2014	Original discovery of IMNEPD1; <i>PTRH2</i> c.269_270delCT (p.Ala90fs) in consanguineous Turkish sibs by homozygosity mapping + WES	Human clinical / genetics
25558065	Alazami 2015	Q85P recurrent variant in Arab consanguineous families	Human clinical / genetics
27129381	Picker-Minh 2016	Q85P destabilizes protein (normal mRNA, reduced protein); genotype–phenotype delineation	Human clinical / in vitro
31057140	Le 2019	p.Trp108* nonsense variant; severe phenotype	Human clinical / genetics
33717719	Rea 2021	Digenic PTRH2+KIF1A neuro-pancreatic syndrome; "NPS" umbrella term	Human clinical / genetics
15006356	Jan 2004	BIT1 released on detachment binds AES/Groucho → caspase-independent anoikis	In vitro / mechanism
41807994	Na 2026	PTRH2–MFN1/2–FKBP8 mitochondrial fission/mitophagy; truncation mutants; amino-acid-deprivation rescue	In vitro / mechanism
12475932	Rosas-Sandoval 2002	Eukaryotic PTH2 hydrolase activity; conservation; non-essential in yeast	In vitro / computational / comparative

Additional supporting sources: Sharkia et al. cohort meta-analyses (*Genes* 2023, PMC10217894; 2024 update PMC11675358) for aggregated phenotype frequencies; the *Cerebellum* 2022 study (doi:10.1007/s12311-022-01488-z) for the Purkinje-conditional knockout mTOR/S6 evidence; and *PNAS Nexus* 2025 for the PTRH2–TRABID–mt-ND5 Complex I link.

Convergence and challenge. The three mechanistic arms are individually supported and converge on the same cellular endpoint (impaired survival of post-mitotic cells), which the mouse models independently confirm at the organ level. A residual tension is whether the

hydrolase-dependent (translational) or hydrolase-independent (signaling/dynamics) functions dominate in vivo; the milder phenotypes of destabilizing missense alleles versus more severe truncations suggest a graded dependence on residual protein rather than a single dominant arm.

Limitations and Knowledge Gaps

- **Extreme rarity (n≈19–32 patients).** Frequency estimates, genotype–phenotype correlations, and prognosis carry wide uncertainty; no formal survival, incidence, or prevalence statistics exist (Orphanet: <1/1,000,000 / unknown).
 - **Ascertainment bias.** Nearly all reported families are consanguineous (Middle Eastern/ North African), so the true global spectrum and milder or compound-heterozygous presentations are likely under-recognized.
 - **Mechanistic integration is partly inferred.** The relative in-vivo contribution of the three molecular arms is not quantified; much dynamics/mitophagy data come from overexpressed truncation mutants in cell lines rather than patient tissue.
 - **No human tissue omics.** No published transcriptomic, proteomic, or metabolomic profiling of patient tissues; molecular profiling relies on fibroblasts and mouse tissue.
 - **Endocrine mechanism gap.** Why islets are relatively spared in mice yet ~40% of patients develop diabetes, and the basis of hypothyroidism, remain incompletely explained.
 - **No natural animal disease** and no large-animal model; the mouse is the sole faithful in-vivo system, and it dies too early to model late endocrine features.
 - **No disease-modifying therapy** and no registered clinical trials; the amino-acid-deprivation rescue is a cell-model observation only.
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** to capture natural history, survival, genotype–phenotype correlations, and endocrine-pancreatic penetrance across non-consanguineous populations.
2. **Patient-derived iPSC models** (cerebellar/Purkinje organoids, pancreatic and Schwann-cell differentiations) to test the three mechanistic arms in disease-relevant human cell types.

3. **Test the amino-acid-deprivation / fusion-promoting rescue** (Na 2026) and mitochondrial-dynamics modulators in patient iPSC neurons and *Ptrh2* mouse models as candidate therapeutic strategies.
4. **Multi-omics of patient and mouse tissue** (single-cell/spatial transcriptomics of cerebellum, nerve, pancreas; proteomics; targeted metabolomics of OXPHOS/ROS markers) to rank the contribution of translation vs signaling vs dynamics arms.
5. **Structure–function dissection** separating hydrolase-dependent from hydrolase-independent alleles (catalytic-dead vs adhesion-signaling-dead mutants) to determine which function is limiting in each affected tissue.
6. **Endocrine focus:** longitudinal beta-cell/thyroid characterization in *Ptrh2* models (using conditional or hypomorphic alleles that survive longer) to explain the diabetes and hypothyroidism seen in patients.
7. **Biomarker standardization:** evaluate fecal elastase-1, fat-soluble vitamin panels, nerve-conduction indices, and cerebellar MRI-volumetry as diagnostic and progression markers.

*Report compiled from 9 confirmed findings across a 5-iteration autonomous investigation. Evidence sources span human clinical case series/cohorts, patient fibroblasts, CRISPR cell lines, constitutive and Purkinje-conditional *Ptrh2* knockout mice, and comparative/computational analyses.*
